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High Performance Processors and Systems

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Name
Last Name

Problem 1 (5 points)	
Problem 2 (4 points)	
Problem 3 (5 points)	
Question 1 (6 points)	
Question 2 (6 points)	
Question 3 (6 points)	
Total (32 points)	

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem has to be correct (at least it has to be evaluated with 85% of the “available” points)
- At least 8 points from the problems
- At least 9 points from the questions

Problem 1

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 100, R1 is set to 0)

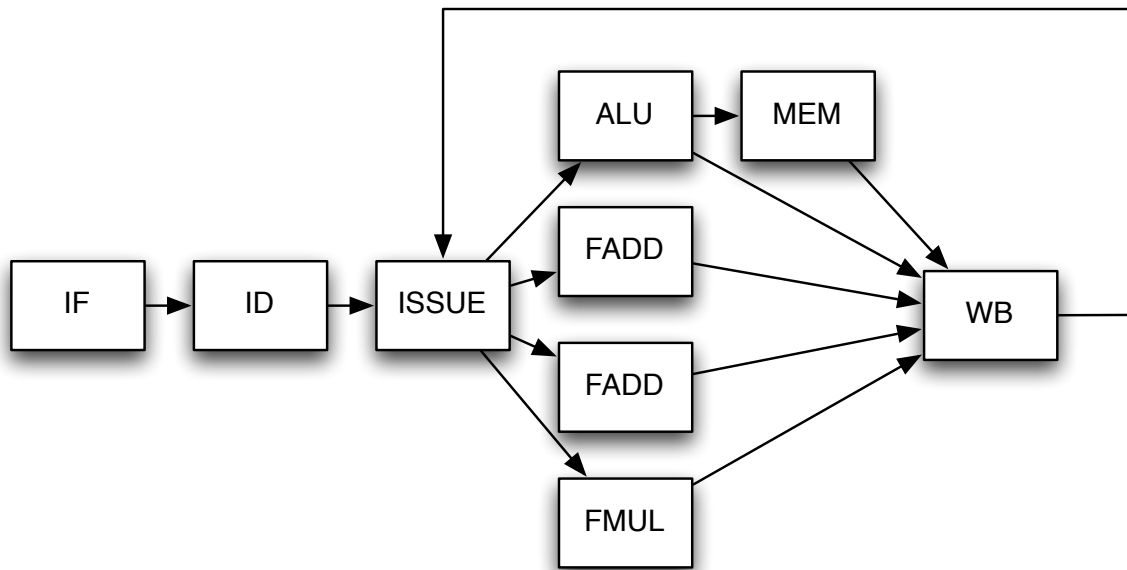
```
LOOP:    LD      F1  0    R0
         ADDD    F2  F1   F1
         ADDI    R1  R1   300
LOOP2:   MULTD   F2  F2   F1
         SUBI    R1  R1   1
         BNEZ    R1  LOOP2
         SUBI    R0  R0   2
         BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 2 cycles (includes time in ALU)
- Floating-point add instructions take 3 cycles
- Floating-point multiply instructions take 4 cycles
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

- **I** - When an instruction enters the issue stage
- **E** - When an instruction starts execution (enters FU)
- **C** - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

I ₁	L.D	F0, 0(R0)
I ₂	MUL.D	F0, F1, F1
I ₃	ADD.D	F2, F1, F1
I ₄	ADD.D	F4, F3, F3
I ₅	ADDI	R1, R3, 8
I ₆	ADD.D	F6, F0, F0

Problem 1.A

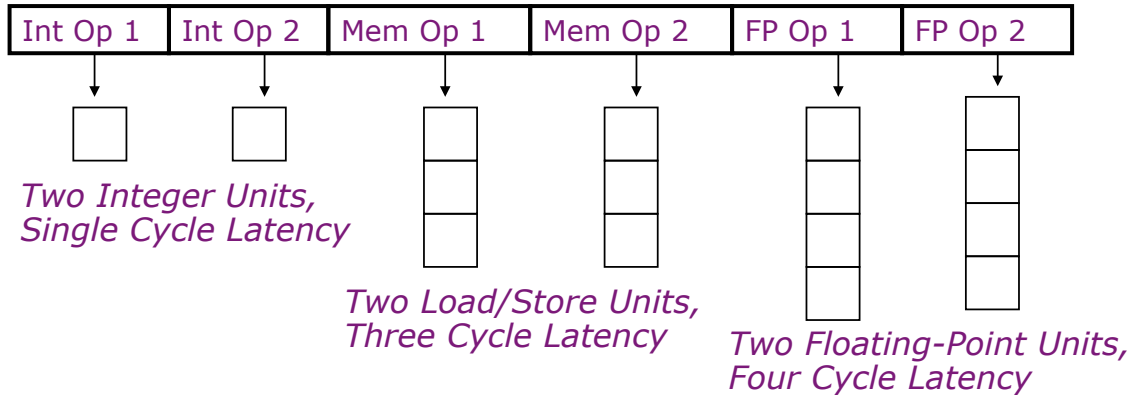
Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

[illegible]

Problem 3

P3.A: Explain the main differences between a VLIW architecture and a superscalar architecture, in terms of instructions scheduling, hardware complexity, and code size.

P3.B: Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
1. Loop:      LD F0, 0(R1)
               FADD F4, F0, F2
               SD F4, 0(R1)
               ADD R1, 8
               BNE R1 R2 Loop
```

How many FLOPS/cycle?

2. With an unroll equal to 4 we can obtain the following code

```
Loop:      LD F0, 0(R1)
            LD F6, 8(R1)
            LD F10, 16(R1)
            LD F14, 24(R1)
            FADD F4, F0, F2
            FADD F8, F6, F2
            FADD F12, F10, F2
            FADD F16, F14, F2
            SD F4, 0(R1)
            SD F8, 8(R1)
            ADD R1, 32
            SD F12, -16(R1)
            BNE R1 R2 Loop
            SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P3.A

Answer P3.B.1

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Answer P3.B.2

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Question 1

1. Explain the main differences between the scoreboard algorithm and Tomasulo scheduling algorithm. Be specific.
2. How WAW and WAR are treated in the two algorithms?

Question 2

Consider a 2 processors systems with shared memory that adopts the MESI write-invalidate write-back protocol.

1. Define which are the states in which each cache block can be and explain how is the coherence ensured by this protocol;
2. Explain what happens under this protocol when the following operations are performed by the two processors P1 and P2, assuming a shared variable stored in address A1 of the memory with initial variable equal to 0. Consider initially that the cache blocks are invalid.
 - a. P1 reads A1
 - b. P2 reads A1
 - c. P2 writes 7 in A1
 - d. P1 reads A1
 - e. P1 writes 3 in A1

Operation	Cache content of P1	Status of the cache block at the end of the operation	Cache content of P2	Status of the cache block at the end of the operation	Memory content for location A1
P1 reads A1					
P2 reads A1					
P2 writes A1					
P1 reads A1					
P1 writes 3 in A1					

Question 3

Multithreading

a. Name three reasons that industry leaders (e.g. Intel) decided around 2002 to stop trying to improve individual processor performance and start producing multicore processors instead.

b. What is Simultaneous Multithreading and which is the difference with fine-grained multithreading?

Simultaneous Multithreading is a technique that allows instructions from multiple threads to exist in the pipeline at the same time. It is term that is applied to superscalar, out-of-order pipelines.

c. What hardware enhancements would be required transform a superscalar, out-of-order processor, into one that performs simultaneous multithreading? Be explicit.

